iMapper: An Interactive Tool for Biomedical Concept Harmonization and Semantic Mapping

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Abstract

Data curation, mapping, and harmonization are foundational activities in the data lifecycle, forming the cornerstone of effective data stewardship and enabling the reuse, interoperability, and reproducibility of research data. In biomedical and clinical research, these activities ensure that data adhere to the FAIR (Findable, Accessible, Interoperable, and Reusable) principles by creating explicit semantic relationships between heterogeneous data sources. Recent developments in large language models (LLMs) and retrieval-augmented generation (RAG) offer new opportunities to transform this landscape providing efficient automated suggestions but they often focus on exact matches and lack mechanisms for reasoning or handling imperfect but valuable correspondences. In practice, human expertise remains critical, as curators must validate mappings, resolve ambiguities, and ensure semantic accuracy.

To address this need, we present an interactive biomedical concept harmonization and semantic mapping tool that combines Retrieval-Augmented Generation (RAG) based mapping with LLM-based reasoning. The system retrieves candidate matches for input entities, infers relationships such as equivalent, subclass or superclass, and generates structured, explainable reasoning traces, allowing curators to review and validate suggestions efficiently. By establishing relationships and mapping concepts to knowledge graphs, the tool enables integration of heterogeneous datasets without losing information from imperfect matches, supporting meaningful knowledge discovery.

Keywords

Data harmonization,, data stewardship, large language models, semantic mapping, RAG1 CEUR-WS

Data mapping and harmonization address the challenge of data silos and fragmented data integration in evidence-based medicine. Directly mapping concepts across data sources constitutes data harmonization, but relying solely on exact or "perfect" matches can lead to information loss when concepts are contextually similar but not identical. To mitigate this, semantic relationships must be established between concepts, enabling features to be integrated based on contextual meaning rather than strict equivalence. With the rise of knowledge graphs and neuro-symbolic AI for explainability, linking cohort-level patient features to biomedical knowledge graphs supports a deeper understanding of biological mechanisms. Moreover, it is often necessary to extend existing knowledge graphs with dataset-specific features using edge/relationships like subclassOf or superClassOf when contextually relevant, making relationship discovery as critical as harmonization.

We are developing a biomedical concept harmonization and semantic mapping tool built around three principles: enabling easy relationship establishment, supporting interactive mapping validation and export, and leveraging LLM-based reasoning. The tool uses a Retrieval-Augmented Generation (RAG) approach where input entities are converted into embeddings and compared to concepts in a knowledge graph to retrieve candidate matches. An LLM then provides structured reason explaining the best match and inferred relationships, such as equivalent, subclass or superclass or None. Curators review and validate suggestions through an interactive interface, ensuring transparency and semantic reliability. Currently, we are testing the system in zero-shot settings using prompt-based reasoning, and on two knowledge graph sources: a causal Alzheimer's knowledge graph (with nodes including pathology, genes, and drugs) developed in-house, and the OMOP vocabulary converted into graph structure.

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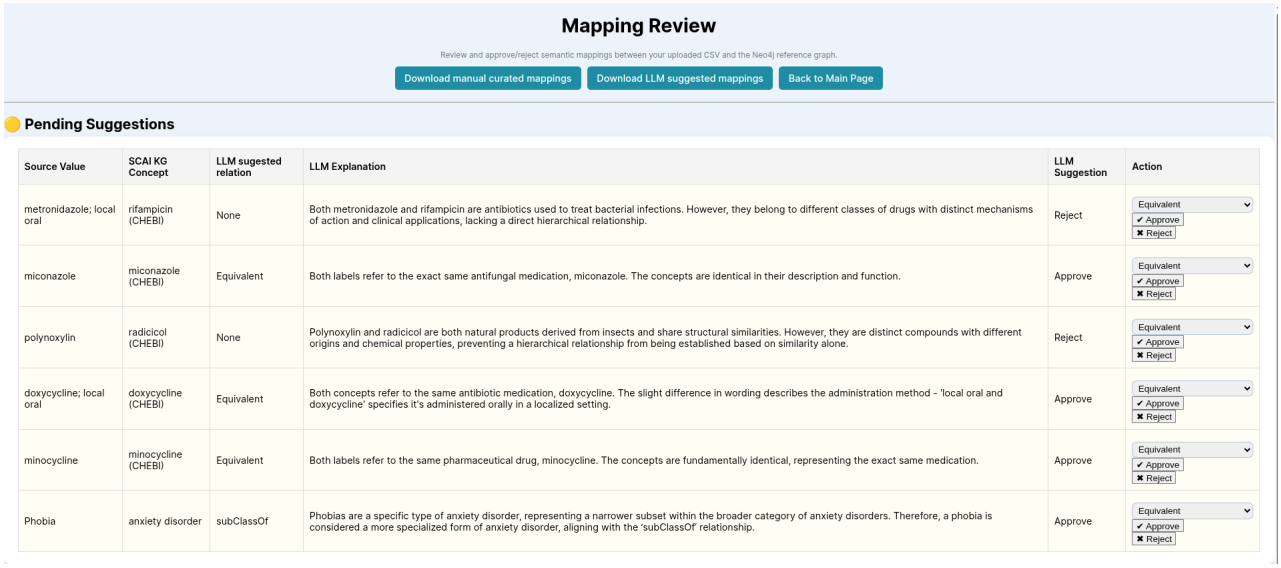
Preliminary results

Initial testing shows that the tool can accurately map biomedical concepts even when exact matches are unavailable, while also inferring meaningful equivalent, subclass or superclass relationships to place novel concepts within the knowledge graph. The LLM-generated reasoning provides clear, explainable justification for each suggestion, enhancing curator confidence. To ensure task adherence, we employ carefully designed system prompts and structured response formats, guiding the LLM to focus on the harmonization task. The Gemma model has delivered particularly promising results in zero-shot settings, producing reliable mappings and relationship inferences without task-specific training. By establishing relationships and mapping dataset features to the knowledge graph, the tool enables effective integration of heterogeneous datasets. When a feature is equivalent to an existing node, it can be added as a property of that node; when it represents a subclass or superclass, it can be introduced as a new node, thereby enriching the knowledge graph and supporting inference. This approach facilitates new knowledge discovery by allowing previously unmapped or ambiguous data to be meaningfully incorporated into the graph.



Source Value	SCAI KG Concept	LLM sugested relation	LLM Explanation	LLM Suggestion	Action
metronidazole; local oral	rifampicin (CHEBI)	None	Both metronidazole and rifampicin are antibiotics used to treat bacterial infections. However, they belong to different classes of drugs with distinct mechanisms of action and clinical applications, lacking a direct hierarchical relationship.	Reject	Equivalent ✔ Approve ✖ Reject
miconazole	miconazole (CHEBI)	Equivalent	Both labels refer to the exact same antifungal medication, miconazole. The concepts are identical in their description and function.	Approve	Equivalent ✔ Approve ✖ Reject
polynoxylin	radicicol (CHEBI)	None	Polynoxylin and radicicol are both natural products derived from insects and share structural similarities. However, they are distinct compounds with different origins and chemical properties, preventing a hierarchical relationship from being established based on similarity alone.	Reject	Equivalent ✔ Approve ✖ Reject
doxycycline; local oral	doxycycline (CHEBI)	Equivalent	Both concepts refer to the same antibiotic medication, doxycycline. The slight difference in wording describes the administration method - 'local oral and doxycycline' specifies it's administered orally in a localized setting.	Approve	Equivalent ✔ Approve ✖ Reject
minocycline	minocycline (CHEBI)	Equivalent	Both labels refer to the same pharmaceutical drug, minocycline. The concepts are fundamentally identical, representing the exact same medication.	Approve	Equivalent ✔ Approve ✖ Reject
Phobia	anxiety disorder	subClassOf	Phobias are a specific type of anxiety disorder, representing a narrower subset within the broader category of anxiety disorders. Therefore, a phobia is considered a more specialized form of anxiety disorder, aligning with the 'subClassOf' relationship.	Approve	Equivalent ✔ Approve ✖ Reject

Figure 1: Overview of interactive mapping interface displaying LLM sugested relation,LLM explanation, and user action to approve or reject the suggested mapping. The tool also allows user to directly download the LLM suggested mappings, or make a decision by themselves.

Declaration on Generative AI

During the preparation of this work, the author(s) used ChatGPT, Grammarly in order to: Grammar and spelling check, and improve writing style. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.